

Chapter 3

Disorders of water metabolism: diabetes insipidus and the syndrome of inappropriate antidiuretic hormone secretion

JOSEPH G. VERBALIS*

Georgetown University, Washington, DC, USA

Disorders of body fluids are among the most commonly encountered problems in the practice of clinical medicine. This is in large part because many different disease states can potentially disrupt the finely balanced mechanisms that control the intake and output of water and solute. Since body water is the primary determinant of the osmolality of the extracellular fluid (ECF), disorders of body water homeostasis can be broadly divided into *hypo-osmolar disorders*, in which there is an excess of body water relative to body solute, and *hyperosmolar disorders*, in which there is a deficiency of body water relative to body solute. Because sodium is the main constituent of plasma osmolality, these disorders are typically characterized by *hyponatremia* and *hypernatremia*, respectively. Before discussing these disorders, this chapter will briefly review the regulatory mechanisms underlying water and sodium metabolism, the two major determinants of body fluid homeostasis.

WATER METABOLISM

Water metabolism represents a balance between the intake and excretion of water. Each side of this balance equation can be considered to consist of a “regulated” and an “unregulated” component, the magnitudes of which can vary quite markedly under different physiologic and pathophysiologic conditions. The unregulated component of water intake consists of the intrinsic water content of ingested foods, the consumption of beverages primarily for reasons of palatability or desired secondary effects (e.g., caffeine), or for social or habitual reasons (e.g., alcoholic beverages), whereas the regulated component of water intake consists of fluids consumed in response to a perceived sensation of thirst. Similarly, the unregulated component of water excretion occurs via insensible water losses from a variety of sources

(cutaneous losses from sweating, evaporative losses in exhaled air, gastrointestinal losses) as well as the obligate amount of water that the kidneys must excrete to eliminate solutes generated by body metabolism, whereas the regulated component of water excretion is composed of the renal excretion of free water in excess of the obligate amount necessary to excrete metabolic solutes (Verbalis, 1997b). In effect, the regulated components are those that act to maintain water balance by compensating for whatever perturbations result from unregulated water losses or gains. Within this framework, it is clear that the two major mechanisms responsible for regulating water metabolism are thirst and pituitary secretion of the hormone vasopressin.

Thirst

Thirst is the body's defense mechanism to increase water consumption in response to perceived deficits of body fluids. Thirst can be stimulated in animals and man either by intracellular dehydration caused by increases in the effective osmolality of the ECF, or by intravascular hypovolemia caused by losses of ECF. Substantial evidence to date has supported mediation of the former by osmoreceptors located in the anterior hypothalamus of the brain, whereas the latter appears to be stimulated primarily via activation of low- and/or high-pressure baroreceptors, with a likely contribution from circulating angiotensin II during more severe degrees of intravascular hypovolemia and hypotension (Fitzsimons, 1992; Stricker and Verbalis, 2013). Controlled studies in animals have consistently reported thresholds for osmotically induced drinking ranging from 1% to 4% increases in plasma osmolality above basal levels; analogous studies in humans using quantitative estimates of subjective symptoms of thirst have confirmed that increases in plasma

*Correspondence to: Dr. Joseph G. Verbalis, 232 Building D, Georgetown University, 4000 Reservoir Rd. NW, Washington, DC 20007, USA. Tel: +1-202-687-2818, Fax: +1-202-687-2040, E-mail: verbalis@georgetown.edu

osmolality of similar magnitudes are necessary to produce an unequivocal sensation described as “thirst” (Robertson, 1983; Thompson et al., 1986).

Conversely, the threshold for producing hypovolemic, or extracellular, thirst is significantly greater in both animals and humans. Studies in several species have shown that sustained decreases in plasma volume or blood pressure of at least 4–8%, and in some species 10–15%, are necessary to consistently stimulate drinking. In humans, it has been difficult to demonstrate any effects of mild to moderate hypovolemia to stimulate thirst independently of osmotic changes occurring with dehydration. This blunted sensitivity to changes in extracellular fluid volume or blood pressure in humans probably represents an adaptation that occurred as a result of the erect posture of primates, which predisposes them to wider fluctuations in blood and atrial filling pressures as a result of orthostatic pooling of blood in the lower body; stimulation of thirst (and vasopressin secretion) by such transient postural changes in blood pressure might lead to overdrinking and inappropriate antidiuresis in situations where the ECF volume was actually normal but only transiently maldistributed. Consistent with a blunted response to baroreceptor activation, studies have also shown that systemic infusion of angiotensin II to pharmacologic levels is a much less potent stimulus to thirst in humans than in animals (Phillips et al., 1985). Nonetheless, this response is not completely absent in humans, as demonstrated by rare cases of polydipsia in patients with pathologic causes of hyper-reninemia.

Although osmotic changes clearly are more effective stimulants of thirst than are volume changes in humans, it is not clear whether relatively small changes in plasma osmolality are responsible for day-to-day fluid intakes. Most humans consume the majority of their ingested water as a result of the unregulated components of fluid intake discussed previously, and generally ingest volumes in excess of what can be considered to be actual “need” (de Castro, 1988). Consistent with this observation is the fact that under most conditions plasma osmolalities in man remain within 1–2% of basal levels, and these relatively small changes in plasma osmolality are generally below the threshold levels that have been found to stimulate thirst in most individuals. This suggests that despite the obvious vital importance of thirst during pathologic situations of hyperosmolality and hypovolemia, under normal physiologic conditions water balance in man is accomplished more by regulated free water excretion than by regulated water intake (Verbalis, 1997b).

Vasopressin secretion

The primary physiologic action of *arginine vasopressin* (AVP) is its function as a water-retaining hormone.

The central sensing system (*osmostat*) for control of release of AVP is located in the hypothalamus anterior to the third ventricle that also includes the circumventricular organ, the *organum vasculosum of the lamina terminalis* (OVLT). The osmostat controls release of AVP to cause water retention, and also stimulates thirst to cause water repletion (Stricker and Verbalis, 2013). Osmotic regulation of AVP release and thirst are usually closely coupled, but experimental lesions and some pathologic situations in humans demonstrate that each can be regulated independently (Baylis and Thompson, 1988). The primary extracellular osmolyte to which the osmoreceptor responds is sodium. Under normal physiologic conditions, glucose and urea readily traverse neuron cell membranes and do not stimulate release of AVP. Basal osmolality in normal subjects lies between 280 and 295 mOsm/kg H₂O, but for each individual osmolality is maintained within narrow ranges. Increases in plasma osmolality of as little as 1–2% will stimulate the osmoreceptors to release AVP. Basal plasma levels of AVP are 0.5–2 pg/mL, which are sufficient to maintain urine osmolality above plasma osmolality and urine volume in the range of 2–3 L/day. When AVP levels are suppressed below 0.5 pg/mL, maximum urine osmolality decreases to less than 100 mOsm/kg H₂O and a free water diuresis ensues to levels approaching 800–1000 mL/hour (18–24 L/day). Increases in plasma osmolality cause a linear increase in plasma AVP and a corresponding linear increase in urine osmolality (Robertson, 1976). At a plasma osmolality of approximately 295 mOsm/kg H₂O, urine osmolality is maximally concentrated to 1000–1200 mOsm/kg H₂O. Thus, the entire physiologic range of urine concentration is accomplished by relatively small changes in plasma AVP of 0–5 pg/mL (Robinson and Verbalis, 2011).

AVP secretion is also stimulated by low blood volume and pressure. High-pressure baroreceptors are located in the aorta and carotid sinus, and low-pressure baroreceptors are located in the right and left atria. Stimuli for pressure and volume receptors are carried via the glossopharyngeal (ninth) and vagal (tenth) cranial nerves to the *nucleus tractus solitarius* in the brainstem. Subsequent secondary and tertiary projections converge on the magnocellular neurons, where they provide inhibitory as well as excitatory inputs. Decreases in blood pressure or vascular volume stimulate AVP release, whereas situations that increase blood volume or left atrial pressure (e.g., negative-pressure breathing) decrease secretion of AVP. The release of AVP in response to changes in volume or pressure is less sensitive than the release in response to osmoreceptors, and generally a 10–15% reduction in blood volume or pressure is needed to stimulate release of AVP. However, once arterial pressure falls below this threshold, the stimulated response is

exponential and plasma levels of AVP achieved that are markedly greater than those achieved by osmotic stimulation (Robertson, 1976). Other nonosmotic stimuli, such as nausea and intestinal traction, also act through similar nonosmotic neural pathways to release AVP.

Vasopressin actions

Three known receptor subtypes mediate the actions of AVP. They all are classic G protein-coupled receptors with seven transmembrane domains, and are classified according to the second messenger system to which they are coupled (Thibonnier et al., 1998). The AVP V1a (V1aR) and V1b (V1bR) receptors are linked to the phosphoinositol signaling pathway via Gq/11 GTP binding proteins that activate phospholipase C activity, with intracellular calcium acting as the second messenger. V1aR are present on vascular smooth muscle cells, hepatocytes, and platelets, and mediate the well-known pressor effects of AVP on peripheral resistance and blood pressure. V1bR are found predominately on corticotroph cells of the anterior pituitary, where they mediate corticotropin (ACTH) release in concert with the well-known effects of corticotropin-releasing hormone (CRH). V2R, or antidiuretic receptors, are mainly localized in the collecting duct cells of the kidney where they regulate water excretion. V2R are G protein-coupled receptors that activate adenyl cyclase with subsequent increased intracellular cyclic AMP levels upon ligand activation. The increased cAMP initiates the movement of *aquaporin-2* (AQP2) water channels from the cytoplasm to the apical (luminal) membrane of the collecting duct cells. Once inserted into the apical membrane, these channels allow facilitated rapid transport of water from the collecting duct lumen into the cell along osmotic gradients (Knepper, 1997). The water then exits the cell through the basolateral membrane and into the kidney medullary circulation via aquaporin-3 and aquaporin-4 water channels, which are constitutively present in the basolateral membrane. This entire process is termed *antidiuresis*. In the absence of AVP, the AQP2 channels are re-internalized from the apical membrane into subapical vesicles. This prevents active reabsorption of water from the collecting duct lumen, resulting in *diuresis*. In addition to this rapid “shuttling” of the AQP2 water channels to regulate water reabsorption on a minute-to-minute basis, AVP also acts via V2R to regulate long-term stores of AQP2; i.e., increased AVP stimulates AQP2 synthesis and the absence of AVP results in decreased AQP2 synthesis (Knepper, 1998). The hypertonic medullary interstitium determines the maximum concentration of the final urine, which is isotonic with the inner medulla of the kidney under conditions of maximal antidiuresis (Knepper, 1997; Nielsen et al., 2002).

Integration of thirst and AVP secretion

A synthesis of what is presently known about the regulation of thirst and AVP secretion in man leads to a relatively simple but elegant system to maintain water balance (Verbalis, 1997b; Stricker and Verbalis, 2013). Under normal physiologic conditions, the sensitivity of the osmoregulatory system for AVP secretion accounts for maintenance of plasma osmolality within narrow limits by adjusting renal water excretion to small changes in osmolality. Stimulated thirst does not represent a major regulatory mechanism under these conditions, and unregulated fluid ingestion supplies adequate water in excess of true “need,” which is then excreted in relation to osmoregulated pituitary AVP secretion. However, when unregulated water intake cannot adequately supply body needs in the presence of plasma AVP levels sufficient to produce maximal antidiuresis, then plasma osmolality rises to levels that stimulate thirst and produce water intake proportional to the elevation of osmolality above this threshold. In such a system thirst essentially represents a backup mechanism called into play when pituitary and renal mechanisms prove insufficient to maintain plasma osmolality within a few percent of basal levels. This arrangement has the advantage of freeing man from frequent episodes of thirst that would require a diversion of activities toward behavior oriented to seeking water when water deficiency is sufficiently mild to be compensated for by renal water conservation, but would stimulate water ingestion once water deficiency reaches potentially harmful levels. Stimulation of AVP secretion at plasma osmolalities below the threshold for subjective thirst acts to maintain an excess of body water sufficient to eliminate the need to drink whenever slight elevations in plasma osmolality occur. This system of differential effective thresholds for thirst and AVP secretion nicely complements many studies that have demonstrated excess unregulated, or “need-free”, drinking in both man and animals (Fitzsimons, 1992).

SODIUM METABOLISM

Maintenance of sodium homeostasis requires a balance between intake and excretion of Na^+ . As in the case of water metabolism, it is possible to define regulated and unregulated components of both Na^+ intake and Na^+ excretion. Unlike water intake, however, there is little evidence in humans to support a significant role for regulated Na^+ intake, with the possible exception of some pathologic conditions. Consequently, there is an even greater dependence on mechanisms for regulated renal excretion of sodium than is the case for excretion of water (Verbalis, 1997a). Whether for this reason or

not, the mechanisms for renal excretion of sodium are more numerous and substantially more complex than the relatively simple, albeit quite efficient, system for AVP-regulated excretion of water.

Salt appetite

The only solute for which any specific appetite has been clearly demonstrated in man is sodium (as with animals, this is generally expressed as an appetite for the chloride salt of sodium, so it is usually called *salt appetite*). Because of the importance of Na^+ for ensuring maintenance of the ECF volume, which in turn directly supports blood volume and pressure, its uniqueness insofar as meriting a specific mechanism for regulated intake seems appropriate. However, despite abundant evidence in many different species demonstrating a salt appetite that is proportionately related to Na^+ losses (Denton, 1982), there is only one pathologic condition in which a specific stimulated sodium appetite has been unequivocally observed in humans, namely Addison's disease caused by adrenal insufficiency. Almost since the initial discovery of this disorder, salt craving has remained one of the well-known manifestations of Addison's disease (Wilkins and Richter, 1940). A robust salt appetite also occurs prominently in adrenalectomized animals, and appears to be related in part to the high plasma levels of adrenocorticotrophic hormone (ACTH) produced as a result of the loss of cortisol feedback on the pituitary. However, despite the presence of Na^+ deficiency in most patients with untreated Addison's disease, only 15–20% of such patients manifest salt-seeking behavior (Orth and Kovacs, 1998). Even more striking is the apparent absence of salt appetite during a variety of other disorders causing severe Na^+ and ECF volume depletion in humans (e.g., patients with hemorrhagic blood loss, diuretic-induced hypovolemia, or hypotension of any etiology become thirsty when intravascular deficits are marked, but almost never express a pronounced desire for salty foods or fluids).

Renal sodium excretion

Although specific mechanisms exist for regulated renal excretion of all major electrolytes, none is as numerous or as complex as those controlling Na^+ excretion, which is not surprising in view of the fact that maintenance of ECF volume is crucial to normal health and function. The most important of these mechanisms are glomerular filtration rate, aldosterone secretion and renal effects, and intrarenal hemodynamic and peritubular factors. In view of the complexity of these mechanisms, the reader is referred to more complete reviews of this topic (Kirchner and Stein, 1994; Reeves and Andreoli, 2001).

HYPO-OSMOLALITY

Hypo-osmolality indicates excess water relative to solute in the ECF; because water moves freely between the ECF and the intracellular fluid (ICF), this also indicates an excess of total body water relative to total body solute. Imbalances between body water and solute can be generated either by *depletion* of body solute more than body water, or by *dilution* of body solute from increases in body water more than body solute. This is an oversimplification of complex physiology, and most hypo-osmolar states include components of both solute depletion and water retention. Nonetheless, this general concept has proven to be useful because it provides a simple framework for understanding the basic etiologies of hypo-osmolar disorders.

Differential diagnosis

Definitive identification of the etiology of hypo-osmolality is not always possible at the time of presentation, but categorization according to the patient's ECF volume status represents the first step in ascertaining the underlying cause of the disorder (Verbalis, 2012).

DECREASED ECF VOLUME (HYPOVOLEMIA)

Clinically detectable hypovolemia indicates some degree of solute depletion. Even isotonic or hypotonic fluid losses can cause hypo-osmolality if water or hypotonic fluids are subsequently ingested or infused. A low urine sodium concentration (U_{Na}) suggests a nonrenal cause of solute depletion, whereas a high U_{Na} suggests renal causes of solute depletion. Diuretic use is the most common cause of hypovolemic hypo-osmolality. Most etiologies of solute losses causing hypovolemic hypo-osmolality will be clinically apparent, although some salt-wasting nephropathies and mineralocorticoid deficiency may be difficult to diagnose during early phases of these diseases.

NORMAL ECF VOLUME (EUVOLEMIA)

Virtually any disorder causing hypo-osmolality can present with a volume status that appears normal by standard methods of clinical evaluation. Because clinical assessment of volume status is not very sensitive, the presence of normal or low blood urea nitrogen and uric acid concentrations are helpful laboratory correlates of relatively normal ECF volume. In these cases, a low U_{Na} (<30 mmol/L) suggests depletion hypo-osmolality secondary to ECF losses with subsequent volume replacement by water or other hypotonic fluids. Such patients may appear euvolemic by the usual clinical parameters used to assess hydration status. A high

U_{Na} (≥ 30 mmol/L) generally indicates a dilutional hypo-osmolality such as the syndrome of inappropriate anti-diuretic hormone secretion (SIADH), the most common cause of euvolemic hypo-osmolality. The clinical criteria necessary to diagnose SIADH remain as initially defined by Bartter and Schwartz in 1967 (Bartter and Schwartz, 1967). Many different disorders are associated with SIADH; these can be divided into four major etiologic groups: tumors, CNS disorders, drug effects, and pulmonary diseases.

INCREASED ECF VOLUME (HYPERVOLEMIA)

Clinically detectable hypervolemia indicates whole body sodium excess, and hypo-osmolality in these patients suggests a relatively decreased intravascular volume and/or pressure leading to water retention as a result of elevated plasma AVP levels and decreased distal delivery of glomerular filtrate to the kidneys. Such patients usually have a low U_{Na} because of secondary hyperaldosteronism, but under certain conditions the U_{Na} may be elevated (e.g., diuretic therapy). Hyponatremia generally does not occur until relatively advanced stages of congestive heart failure, cirrhosis, or the nephrotic syndrome, by which time diagnosis is usually not difficult. Renal failure can also cause retention of both sodium and water.

Clinical manifestations

The clinical manifestations of hyponatremia are largely neurologic, and primarily reflect brain edema resulting from osmotic water shifts into the brain (Verbalis, 2010). These range from nonspecific symptoms such as headache and confusion to more severe manifestations such as decreased sensorium, coma, seizures, and death. Significant central nervous system (CNS) symptoms generally do not occur until the serum sodium concentration ($[Na^+]$) falls below 125 mmol/L, and the severity of symptoms can be roughly correlated with the degree of hypo-osmolality. Individual variability is marked, and for any patient the level of serum $[Na^+]$ at which symptoms will appear cannot be accurately predicted. Several factors other than the severity of the hypo-osmolality also affect the degree of neurologic dysfunction. The most important is the time course over which hypo-osmolality develops. Rapid development of severe hypo-osmolality frequently causes marked neurologic symptoms, whereas gradual development over several days or weeks is often associated with relatively mild symptomatology despite profound degrees of hypo-osmolality. This is because the brain counteracts osmotic swelling by extruding extracellular and intracellular solutes, including potassium and a variety of small organic molecules (amino acids, polyols, and methylamines)

called *organic osmolytes* (Gullans and Verbalis, 1993). Since this is a time-dependent process, rapid development of hypo-osmolality can result in brain edema before this adaptation occurs, but with slower development of the same degree of hypo-osmolality brain cells can lose solute sufficiently rapidly to prevent cell swelling, brain edema, and neurologic dysfunction (Verbalis, 2010). Underlying neurologic disease also affects the level of hypo-osmolality at which CNS symptoms appear; moderate hypo-osmolality is of little concern in an otherwise healthy patient, but can cause morbidity in a patient with an underlying seizure disorder. Non-neurologic metabolic disorders (hypoxia, hypercapnia, acidosis, hypercalcemia, etc.) similarly can affect the level of osmolality at which CNS symptoms occur.

Therapy

Correction of hyponatremia is associated with markedly improved neurologic outcomes in patients with severely symptomatic hyponatremia. In a retrospective review of patients who presented with severe neurologic symptoms and serum $[Na^+] < 125$ mmol/L, prompt therapy with isotonic or hypertonic saline resulted in a correction in the range of 20 mmol/L over several days and neurologic recovery in almost all cases. In contrast, in patients who were treated with fluid restriction alone, there was very little correction over the study period (< 5 mmol/L over 72 hours), and the neurologic outcomes were much worse, with most of these patients either dying or entering a persistently vegetative state (Ayus, 1986). Consequently, based on this and many similar retrospective analyses, prompt therapy to rapidly increase the serum $[Na^+]$ represents the standard of care for treatment of patients presenting with severe life-threatening symptoms of hyponatremia.

Chronic hyponatremia is much less symptomatic as a result of the process of brain volume regulation. Because of this adaptation process, chronic hyponatremia is arguably a condition that clinicians feel they may not need to be as concerned about, which has been reinforced by the common usage of the descriptor "*asymptomatic hyponatremia*" for many such patients. However, it is clear that many such patients very often do have neurologic symptoms, even if milder and more subtle in nature, including headaches, nausea, mood disturbances, depression, difficulty concentrating, slowed reaction times, unstable gait, increased falls, confusion, and disorientation (Renneboog et al., 2006). Consequently, all patients with hyponatremia who manifest any neurologic symptoms that could possibly be related to the hyponatremia should be considered as potential candidates for treatment of the hyponatremia, regardless of the chronicity of the hyponatremia or the level of serum $[Na^+]$.

Currently available therapies for treatment of hyponatremia

Conventional management strategies for hyponatremia range from saline infusion and fluid restriction to pharmacologic measures to adjust fluid balance. Consideration of treatment options should always include an evaluation of the benefits as well as the potential toxicities of any therapy, and must be individualized for each patient (Verbalis, 2013; Verbalis et al., 2013). It should always be remembered that sometimes simply stopping treatment with an agent that is associated with hyponatremia is sufficient to reverse a low serum $[\text{Na}^+]$.

ISOTONIC SALINE

The treatment of choice for depletion hyponatremia (i.e., hypovolemic hyponatremia) is isotonic saline ($[\text{Na}^+] = 154 \text{ mmol/L}$) to restore ECF volume and ensure adequate organ perfusion. This initial therapy is appropriate for patients who either have clinical signs of hypovolemia, or in whom a spot U_{Na} is $< 30 \text{ mmol/L}$. However, this therapy is ineffective for dilutional hyponatremias such as SIADH (Schwartz et al., 1957), and continued inappropriate administration of isotonic saline to a euvolemic patient may worsen their hyponatremia (Steele et al., 1997) and/or cause fluid overload.

HYPERTONIC SALINE

Acute hyponatremia presenting with severe neurologic symptoms is life-threatening, and should be treated promptly with hypertonic solutions, typically 3% NaCl ($[\text{Na}^+] = 513 \text{ mmol/L}$), as this represents the most reliable method to quickly raise the serum $[\text{Na}^+]$. A continuous infusion of hypertonic NaCl is usually utilized in inpatient settings. Various formulae have been suggested for calculating the initial rate of infusion of hypertonic solutions (Adroge and Madias, 2000b), but until now there has been no consensus regarding optimal infusion rates of 3% NaCl. One of the simplest methods to estimate an initial 3% NaCl infusion rate utilizes the following relationship (Verbalis, 2013; Verbalis et al., 2013):

$$\begin{aligned} &\text{Patients weight (kg)} \\ &\times \text{desired correction rate (mEq/L/hour)} \\ &= \text{infusion rate of 3\% NaCl (mL/hour)} \end{aligned}$$

Depending on individual hospital policies, the administration of hypertonic solutions may require special considerations (e.g., placement in the intensive care unit, sign-off by a consultant, etc.).

An alternative option for more emergent situations is administration of a 100 mL bolus of 3% NaCl, repeated 1–2 times every 30 min if there is no clinical

improvement, which has been recommended by a consensus conference organized to develop guidelines for prevention and treatment of exercise-induced hyponatremia, an acute and potentially lethal condition (Hew-Butler et al., 2008). Injecting this amount of hypertonic saline intravenously raises the serum $[\text{Na}^+]$ by an average of 2–4 mmol/L, which is well below the recommended maximal daily rate of change of 10–12 mmol/24 hours or 18 mmol/48 hours (Sterns et al., 1994). Because the brain can only accommodate an average increase of approximately 7–8% in brain volume before herniation occurs, quickly increasing the serum $[\text{Na}^+]$ by as little as 2–4 mmol/L in acute hyponatremia can effectively reduce brain swelling and intracranial pressure (Battison et al., 2005).

Many physicians are hesitant to use hypertonic saline in patients with chronic hyponatremia, because it can cause an overly rapid correction of serum sodium levels that can lead to the *osmotic demyelination syndrome* (ODS) (Sterns et al., 1986). Nonetheless, this remains the treatment of choice for patients with severe neurologic symptoms, even when the time course of the hyponatremia is nonacute or unknown.

FLUID RESTRICTION

For patients with chronic hyponatremia, fluid restriction has been the most popular and most widely accepted treatment. When SIADH is present, fluids should generally be limited to 500–1000 mL/24 hours. Because fluid restriction increases the serum $[\text{Na}^+]$ largely by under-replacing the excretion of fluid by the kidneys, some have advocated an initial restriction to 500 mL less than the 24 hour urine output (Robertson, 2006). When instituting a fluid restriction, it is important for the nursing staff and the patient to understand that this includes *all* fluids that are consumed, not just water. Generally the water content of ingested food is not included in the restriction because this is balanced by insensible water losses (perspiration, exhaled air, feces, etc.), but caution should be exercised with foods that have high fluid concentrations (such as fruits and soups). Restricting fluid intake can be effective when properly applied and managed in selected patients, but serum $[\text{Na}^+]$ is increased only slowly (1–2 mmol/L/day) even with severe restriction (Schwartz et al., 1957). In addition, this therapy is often poorly tolerated because of an associated increase in thirst leading to poor compliance with long-term therapy. However, it is economically favorable, and some patients do respond well to this option.

Fluid restriction should not be used with hypovolemic patients, and is particularly difficult to maintain in patients with very elevated urine osmolalities secondary to high AVP levels; in general, if the sum of urine Na^+

and K^+ exceeds the serum $[Na^+]$, most patients will not respond to a fluid restriction since an electrolyte-free water clearance will be difficult to achieve (Furst et al., 2000; Berl, 2008; Decaux, 2009). In addition, fluid restriction is not practical for some patients, particularly patients in intensive care settings, who often require administration of significant volumes of fluids as part of their treatment.

DEMECLOCYCLINE

Demeclocycline, a tetracycline antibiotic, inhibits adenylyl cyclase activation after AVP binds to the V2R in the kidney, and thus targets the underlying pathophysiology of SIADH. This therapy is typically used when patients find severe fluid restriction unacceptable and the underlying disorder cannot be corrected. However, demeclocycline is not approved by the US Food and Drug Administration (FDA) or the European Medicines Agency (EMA) to treat hyponatremia, and can cause nephrotoxicity in patients with heart failure and cirrhosis, although this is usually reversible (Singer and Rotenberg, 1973).

MINERALOCORTICOIDS

Administration of mineralocorticoids, such as fludrocortisone, has been shown to be useful in a small number of elderly patients (Ishikawa et al., 1996). However, the initial studies of SIADH did not show it to be of benefit in patients with SIADH, and it carries the risk of fluid overload and hypertension. Consequently, it is rarely used as primary therapy for hyponatremia.

UREA

Administration of urea has been successfully used to treat hyponatremia because it induces osmotic diuresis and augments free water excretion. Effective doses of urea for treatment of hyponatremia are 30–90 g daily in divided doses (Decaux and Genette, 1981). Unfortunately, its use is limited because there is no US Pharmacopeia (USP) formulation for urea, it is not currently approved by any regulatory agency for treatment of hyponatremia, and there are limited data to support its long-term use. Furthermore, urea is associated with poor palatability leading to patient compliance problems. However, patients with feeding tubes may be excellent candidates for urea therapy since palatability is not a concern, and the use of fluid restriction may be difficult in some patients with high obligate intake of fluids as part of their nutritional and medication therapy. Although mild azotemia can occur with urea therapy, this rarely reaches clinically significant levels.

FUROSEMIDE AND NaCl

The use of furosemide (20–40 mg/day) coupled with a high salt intake (200 mmol/day), which represents an extension of the treatment of acute symptomatic hyponatremia to the chronic management of euvoletic hyponatremia, has also been reported to be successful in selected cases (Decaux et al., 1981). However, the long-term efficacy and safety of this approach is unknown.

ARGININE VASOPRESSIN RECEPTOR (AVPR) ANTAGONISTS

Clinicians have used all of the above conventional therapies for hyponatremia over the past decades. However, conventional therapies for hyponatremia, although effective in specific circumstances, are suboptimal for many different reasons, including variable efficacy, slow responses, intolerable side-effects, and serious toxicities. But perhaps the most striking deficiency of most conventional therapies is that, with the exception of demeclocycline, these therapies do not directly target the underlying cause of almost all dilutional hyponatremias, namely inappropriately elevated plasma AVP levels. A new class of pharmacologic agents, *vasopressin receptor antagonists*, also called “vaptans”, which directly block AVP-mediated receptor activation, have recently been developed for clinical use (Thibonnier et al., 2001; Greenberg and Verbalis, 2006).

Conivaptan and tolvaptan are competitive receptor antagonists of the AVP V2R and have been approved by the FDA for the treatment of euvoletic and hypervolemic hyponatremia and by the EMA for treatment of hyponatremia caused by SIADH. These agents compete with AVP/ADH for binding at its site of action in the kidney, thereby blocking the antidiuresis caused by elevated AVP levels and directly attacking the underlying pathophysiology of dilutional hyponatremia. AVPR antagonists produce electrolyte-free water excretion (called “aquaresis”) without affecting renal sodium and potassium excretion (Ohnishi et al., 1993). The overall result is a reduction in body water without natriuresis, which leads to an increase in the serum $[Na^+]$. One of the major benefits of this class of drugs is that serum $[Na^+]$ is significantly increased by an average of 4–8 mmol/L within 24–48 hours (Schrier et al., 2006; Zeltser et al., 2007); this is considerably faster than the effects of fluid restriction, which can take many days. Also, compliance has not been shown to be problem for vaptans, whereas this is a major problem with attempted long-term use of fluid restriction.

Conivaptan is FDA approved for euvoletic and hypervolemic hyponatremia in hospitalized patients.

It is available only as an intravenous preparation, and is given as a 20 mg loading dose over 30 min, followed by a continuous infusion of 20 or 40 mg (Li-Ng and Verbalis, 2010). Generally, the 20 mg continuous infusion is used for the first 24 hours to gauge the initial response. If the correction of serum $[\text{Na}^+]$ is felt to be inadequate (e.g., < 5 mmol/L), then the infusion rate can be increased to 40 mg/day. Therapy is limited to a maximum duration of 4 days because of drug-interaction effects with other agents metabolized by the CYP3A4 hepatic isoenzyme. Importantly, for conivaptan and all other vaptans, it is critical that the serum $[\text{Na}^+]$ concentration is measured frequently during the active phase of correction of the hyponatremia (a minimum of every 6–8 hours for conivaptan, but more frequently in patients with risk factors for development of the osmotic demyelination syndrome (ODS), such as severely low serum $[\text{Na}^+]$, malnutrition, alcoholism and hypokalemia (Verbalis et al., 2013). If the correction approaches 12 mmol/L in the first 24 hours, the infusion should be stopped and the patient monitored on a fluid restriction. If the correction exceeds 12 mmol/L, consideration should be given to administering sufficient water, either orally or as intravenous D₅W to bring the overall correction below 12 mEq/L. The maximum correction limit should be reduced to 8 mEq/L over the first 24 hours in patients with risk factors for development of ODS mentioned previously. The most common adverse effects include infusion-site reactions, which are generally mild and usually do not lead to treatment discontinuation, headache, thirst, and hypokalemia (Zeltser et al., 2007).

Tolvaptan, an oral AVPR antagonist, is FDA approved for treatment of dilutional hyponatremias. In contrast to conivaptan, oral administration allows it to be used for both short- and long-term treatment of hyponatremia (Schrier et al., 2006). Similar to conivaptan, tolvaptan treatment must be initiated in the hospital so that the rate of correction can be monitored carefully. Patients with a serum $[\text{Na}^+] < 125$ mmol/L are eligible for therapy with tolvaptan as primary therapy; if the serum $[\text{Na}^+]$ is ≥ 125 mmol/L, tolvaptan therapy is only indicated if the patient has symptoms that could be attributable to the hyponatremia and the patient is resistant to attempts at fluid restriction (Otsuka Pharmaceutical Co., 2009). The starting dose of tolvaptan is 15 mg on the first day, and the dose can be titrated to 30 mg and 60 mg at 24 hour intervals if the serum $[\text{Na}^+]$ remains < 135 mEq/L or the increase in serum $[\text{Na}^+]$ has been ≤ 5 mmol/L in the previous 24 hours. As with conivaptan, it is essential that the serum $[\text{Na}^+]$ concentration is measured frequently during the active phase of correction of the hyponatremia (a minimum of every 6–8 hours, but more frequently in patients with risk factors

for development of ODS). Limits for safe correction of hyponatremia and methods to compensate for overly rapid corrections are the same as described previously for conivaptan. One additional factor that helps to avoid overly rapid correction with tolvaptan is the recommendation that fluid restriction not be used during the active phase of correction, thereby allowing the patient's thirst to compensate for an overly vigorous aquaresis. Common side-effects include dry mouth, thirst, increased urinary frequency, dizziness, nausea and orthostatic hypotension, which were relatively similar between placebo and tolvaptan groups in clinical trials (Schrier et al., 2006; Otsuka Pharmaceutical Co., 2009).

Because inducing increased renal fluid excretion via either a diuresis or an aquaresis can cause or worsen hypotension in patients with hypovolemic hyponatremia, vaptans are contraindicated in this patient population (Verbalis et al., 2013). However, clinically significant hypotension was not observed in either the conivaptan or tolvaptan clinical trials in euvolemic and hypervolemic hyponatremic patients. Although vaptans are not contraindicated with decreased renal function, these agents generally will not be effective if the serum creatinine is > 3.0 mg/dL. By virtue of their solubility properties, the vaptans are readily absorbed through the gastrointestinal tract, and therefore likely cross the blood–brain barrier as well, but alterations of central nervous system (CNS) functions have not been observed to date with clinical use of vaptans. This may be due to the fact that most of the CNS effects of AVP have been attributed to V_{1a}R, with no evidence to support the presence or biologic function of V₂R in the CNS (Thibonnier et al., 1998); however, such effects have not been reported even with clinical use of conivaptan, a combined V_{1a}R/V₂R antagonist. Therefore, until more directed and sensitive studies are done to assess potential CNS effects of AVP receptor antagonists, it must be concluded that such effects are absent, or minimal.

Hyponatremia treatment guidelines

Although various authors have published recommendations for the treatment of hyponatremia (Adrogué and Madias, 2000b; Ellison and Berl, 2007; Verbalis et al., 2013; Sterns et al., 2009; Verbalis, 2009), no standardized treatment algorithms have yet been widely accepted. A synthesis of existing expert recommendations for treatment of hyponatremia is illustrated in Figure 3.1. This algorithm is based primarily on the symptomatology of hyponatremic patients, rather than the serum $[\text{Na}^+]$ or the chronicity of the hyponatremia, the latter being often difficult to ascertain.

A careful neurologic history and assessment should always be done to identify potential causes of the

Hyponatremia treatment algorithm

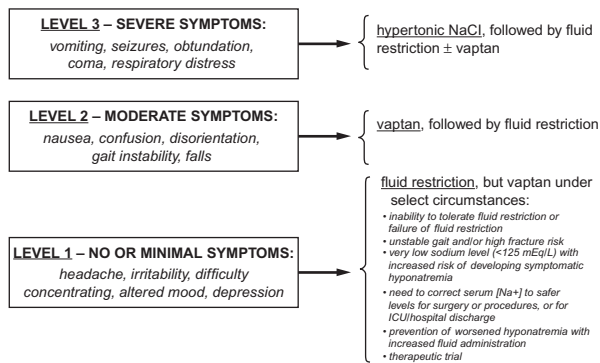


Fig. 3.1. Proposed treatment algorithm for hyponatremia based on degree of symptomatology.

patient's symptoms other than hyponatremia, although it will not always be possible to exclude an additive contribution from the hyponatremia to an underlying neurologic condition. In this algorithm, patients are divided into three groups based on their presenting symptoms: *severe symptoms* (seizures, coma, respiratory arrest, obtundation, and vomiting; these symptoms usually imply a more acute onset or worsening of hyponatremia requiring immediate active treatment); *moderate symptoms* (nausea, confusion, disorientation, and altered mental status; these symptoms may be either chronic or acute, but allow time to elaborate a more deliberate approach to treatment); *mild symptoms* (headache, irritability, inability to concentrate, altered mood, and depression, or a virtual absence of discernible symptoms; these symptoms usually indicate that the patient may have chronic or slowly evolving hyponatremia) (Verbalis, 2012).

Patients with severe symptoms should be treated with hypertonic saline as first-line therapy, followed by fluid restriction with or without AVPR antagonist therapy. Patients with moderate symptoms will benefit from a regimen of vaptan therapy or limited hypertonic saline administration followed by fluid restriction or long-term vaptan therapy. Although moderate neurologic symptoms can indicate that a patient is in an early stage of acute hyponatremia, they more often indicate a chronically hyponatremic state with sufficient brain volume adaptation to prevent marked symptomatology from cerebral edema. Regardless, close monitoring of these patients in a hospital setting is warranted until the symptoms improve or stabilize. Patients with no or minimal symptoms should be managed initially with fluid restriction, although treatment with vaptans may be appropriate for a wide range of specific clinical conditions, foremost of which is a failure to improve the serum $[\text{Na}^+]$ despite reasonable attempts at fluid restriction.

A special case is when spontaneous correction of hyponatremia occurs at an undesirably rapid rate as a result of the onset of a water diuresis. This can occur following cessation of desmopressin therapy in a patient who has become hyponatremic, replacement of glucocorticoids in a patient with adrenal insufficiency, replacement of solutes in a patient with diuretic-induced hyponatremia, or spontaneous resolution of transient SIADH. Brain damage from ODS can clearly ensue in this setting if the preceding period of hyponatremia has been of sufficient duration (usually ≥ 48 hours) to allow brain volume regulation to occur. If the previously discussed correction parameters have been exceeded and the correction is proceeding more rapidly than planned (usually because of continued excretion of hypotonic urine), the pathologic events leading to demyelination can be reversed by administration of hypotonic fluids and desmopressin. Efficacy of this approach is suggested both from animal studies (Soupart et al., 1994) and from case reports in humans (Goldschmidt and Iliescu, 2000; Sterns et al., 2009), even when patients are overtly symptomatic (Oya et al., 2001).

Although this algorithm is based on presenting symptoms at the time of initial evaluation, it should be remembered that in some cases patients initially exhibit more moderate symptoms because they are in the early stages of hyponatremia. In addition, some patients with minimal symptoms are prone to develop more symptomatic hyponatremia during periods of increased fluid ingestion. In support of this, approximately 70% of 31 patients presenting to a university hospital with symptomatic hyponatremia and a mean serum $[\text{Na}^+]$ of 119 mmol/L had pre-existing “asymptomatic” hyponatremia as the most common risk factor identified (Bissram et al., 2007). Consequently, therapy of hyponatremia should also be considered to prevent progression from lower to higher levels of symptomatic hyponatremia, particularly in patients with a past history of repeated presentations for symptomatic hyponatremia.

Monitoring the serum $[\text{Na}^+]$ in hyponatremic patients

The frequency of serum $[\text{Na}^+]$ monitoring is dependent on both the severity of the hyponatremia and the therapy chosen. In all hyponatremic patients neurologic symptomatology should be carefully assessed very early in the diagnostic evaluation to ascertain the symptomatic severity of the hyponatremia and to determine whether the patient requires more urgent therapy. All patients undergoing active treatment with hypertonic saline for symptomatic hyponatremia should have frequent monitoring of serum $[\text{Na}^+]$ and ECF volume status (every 2–4 hours) to ensure that the serum $[\text{Na}^+]$ does not exceed

the recommended levels during the active phase of correction (Verbalis et al., 2013), since overly rapid correction of serum sodium can cause damage to the myelin sheath of nerve cells, resulting in ODS (Sterns et al., 1986). Patients treated with vaptans for mild to moderate symptoms should have serum $[\text{Na}^+]$ monitored every 6–8 hours during the active phase of correction, which will generally be the first 24–48 hours of therapy. Active treatment with hypertonic saline or vaptans should be stopped when the patient's symptoms are no longer present, a safe serum $[\text{Na}^+]$ (usually > 120 mmol/L) has been achieved, or the rate of correction has reached 12 mmol/L within 24 hours or 18 mmol/L within 48 hours (Sterns et al., 1994; Verbalis et al., 2013). Importantly, ODS has not yet been reported either in clinical trials or with therapeutic use of any vaptan as monotherapy to date. In patients with a stable level of serum $[\text{Na}^+]$ treated with fluid restriction or therapies other than hypertonic saline, measurement of serum $[\text{Na}^+]$ daily is generally sufficient, since levels will not change that quickly in the absence of active therapy or large changes in fluid intake or administration.

Long-term treatment of chronic hyponatremia

Some patients will benefit from continued treatment of hyponatremia following discharge from the hospital. In many cases, this will consist of a continued fluid restriction. However, as discussed previously, long-term compliance with this therapy is poor due to the increased thirst that occurs with more severe degrees of fluid restriction. Thus, for selected patients who have responded to tolvaptan in the hospital, consideration should be given to continuing the treatment as an outpatient after discharge. In patients with established chronic hyponatremia, tolvaptan has been shown to be effective at maintaining a normal $[\text{Na}^+]$ for as long as 3 years of continued daily therapy (Berl et al., 2010). However, many patients with inpatient hyponatremia have a transient form of SIADH, without need for long-term therapy. Selection of which patients with inpatient hyponatremia are candidates for long-term therapy should be based on the etiology of the SIADH. In all cases, consideration should be given to a trial of stopping the drug 2–4 weeks following discharge to see if hyponatremia is still present. A reasonable period of tolvaptan cessation to evaluate the presence of continued SIADH is 7 days, since this period was sufficient for demonstration of a recurrence of hyponatremia in the tolvaptan SALT trials (Schrier et al., 2006; Berl et al., 2010). Serum $[\text{Na}^+]$ should be monitored every 2–3 days following cessation of tolvaptan so that the drug can be resumed as quickly as possible in those patients with recurrent hyponatremia,

since the longer the patient is hyponatremic the greater the risk of subsequent ODS with overly rapid correction of the low serum $[\text{Na}^+]$.

HYPEROSMOLALITY

Hyperosmolality indicates a deficiency of water relative to solute in the ECF. Because water moves freely between the ICF and ECF, this also indicates a deficiency of total body water relative to total body solute. Although hypernatremia can be caused by an excess of body sodium, the vast majority of cases are due to losses of body water in excess of body solutes, caused by either insufficient water intake or excessive water excretion. Consequently, most of the disorders causing hyperosmolality are those associated with inadequate water intake and/or deficient pituitary AVP secretion. Although hyperosmolality from inadequate water intake is seen frequently in clinical practice, this is usually not due to an underlying defect in thirst but rather results from a generalized incapacity to obtain and/or ingest fluids, often stemming from a depressed sensorium.

Etiologies and diagnosis

Evaluation of the patient's ECF volume status is important as a guide to fluid replacement therapy, but is not as useful for differential diagnosis since most hyperosmolar patients will manifest some degree of hypovolemia. Rather, assessment of urinary concentrating ability provides the most useful data with regard to the type of disorder present. Using this approach, disorders of hyperosmolality can be categorized as those in which renal water conservation mechanisms are intact but are unable to compensate for inadequately replaced losses of hypotonic fluids from other sources, or those in which renal concentrating defects are a contributing factor to the deficiency of body water (Verbalis, 2012).

Diabetes insipidus

Diabetes insipidus (DI) can result from either inadequate AVP secretion (*central or neurogenic DI*) or inadequate renal response to AVP (*nephrogenic DI*). Central DI is caused by a variety of acquired or congenital anatomic lesions that disrupt the neurohypophysis, including pituitary surgery, tumors, trauma, hemorrhage, thrombosis, infarction, or granulomatous disease (Robertson, 1995), as well as less commonly by genetic mutations of the AVP gene (Babey et al., 2011). Severe nephrogenic DI is most commonly congenital due to defects in the gene for the AVP V2R (X-linked recessive pattern of inheritance) or in the gene for the AQP2 water channel (autosomal recessive pattern of inheritance) (Fujiwara and Bichet, 2005), but relief of chronic urinary obstruction or therapy with

drugs such as lithium can cause an acquired form sufficient to warrant treatment. Acquired nephrogenic DI can result from hypokalemia or hypercalcemia, but the mild concentrating defect generally does not by itself cause hypertonicity and responds to correction of the underlying disorder (Khanna, 2006). Regardless of the etiology of the DI, the end result is a water diuresis due to an inability to concentrate urine appropriately.

Because patients with DI do not have impaired urine Na^+ conservation, the ECF volume is generally not markedly decreased and regulatory mechanisms for maintenance of osmotic homeostasis are primarily activated: stimulation of thirst and pituitary AVP secretion (to whatever degree the neurohypophysis is still able to secrete AVP). In cases where AVP secretion is totally absent (complete DI), patients are dependent entirely on water intake for maintenance of water balance. However, in cases where some residual capacity to secrete AVP remains (partial DI), plasma osmolality can eventually reach levels that allow moderate degrees of urinary concentration. Although untreated DI can lead to both hyperosmolality and volume depletion, until the water losses become severe, volume depletion is minimized by osmotic shifts of water from the ICF into the more osmotically concentrated ECF (Robinson and Verbalis, 2011).

OSMORECEPTOR DYSFUNCTION

The primary osmoreceptors that control AVP secretion and thirst are located in the anterior hypothalamus, and lesions of this region in animals cause hyperosmolality through a combination of impaired thirst and osmotically stimulated AVP secretion (Johnson and Buggy, 1978). Initial reports in humans described this syndrome as “essential hypernatremia,” and subsequent studies used the term “*adipsic hypernatremia*” in recognition of the profound thirst deficits found in most of the patients. All of these syndromes are now grouped together as *disorders of osmoreceptor function* (Baylis and Thompson, 1988). Most of the cases reported to date have represented various degrees of osmoreceptor destruction associated with different brain lesions (Baylis and Thompson, 2001). In contrast to lesions causing central DI, these lesions usually occur more rostrally in the hypothalamus. For all cases of osmoreceptor dysfunction it is important to remember that afferent pathways from the brainstem to the hypothalamus generally remain intact; therefore, these patients will usually have normal AVP and renal concentrating responses to baroreceptor-mediated stimuli such as hypovolemia and hypotension.

DIFFERENTIAL DIAGNOSIS

Distinguishing between central and nephrogenic DI in a patient who is already hyperosmolar entails simply

evaluating the response to a trial of AVP or desmopressin. Administration of AVP (5 units subcutaneously) or, preferably, the selective AVP V2R agonist desmopressin (2 μg subcutaneously or intravenously), should cause a significant increase in urine osmolality within 1–2 hours after injection in patients with central DI, indicating insufficient endogenous AVP secretion. An absent or suboptimal response suggests renal resistance to AVP effects and, therefore, nephrogenic DI. Although conceptually simple, interpretational difficulties often arise because the water diuresis produced by AVP deficiency causes a downregulation of AQP2 synthesis along with a wash-out of the renal medullary concentrating gradient, such that increases in urine osmolality in response to administered AVP or desmopressin are not as great as would be expected (see Table 3.1, the interpretation of urine concentration after AVP/desmopressin).

Because patients with DI generally have an intact thirst mechanism, such patients often do not present with hyperosmolality, but rather have normal plasma osmolality and serum sodium levels with polyuria and polydipsia (Robertson, 1995). In these cases, a fluid deprivation test should be performed in order to raise the serum osmolality and confirm the diagnosis of DI (see Table 3.1 for the procedure and for the interpretation of a fluid deprivation test).

When a diagnosis of central DI is made, magnetic resonance imaging (MRI) of the hypothalamus and neurohypophyseal tract is mandatory to rule out a neoplasm or granulomatous disease as an etiology. In individuals with a normal posterior pituitary, the presence of a pituitary “bright spot” (i.e., a hyperintense signal in the absence of contrast administration, representing the AVP-containing neurosecretory granules) is usually seen on T1-weighted noncontrast sagittal images. Conversely, in patients with central DI, the bright spot is usually absent. However, this test is not definitive, since the pituitary bright spot decreases with age and with disorders that cause dehydration, and it can be present in up to 5% of patients with DI due to the presence of pituitary oxytocin, which also leads to a hyperintense signal on T1-weighted imaging (Robinson and Verbalis, 2011).

Evaluation of anterior pituitary function should be performed in all patients with central DI, especially if glucocorticoid administration and/or replacement unmasks underlying DI. Adrenal insufficiency can cause hypersecretion of AVP, which may be due in part to reductions in systemic blood pressure and cardiac output caused by cortisol deficiency, thereby stimulating pituitary AVP release. Cortisol deficiency is also known to cause increased AVP release from the median eminence into the pituitary portal circulation in an attempt to increase ACTH secretion via effects at pituitary

Table 3.1

Fluid deprivation test for the diagnosis of diabetes insipidus**Procedure**

1. Initiation of the deprivation period depends on the severity of the diabetes insipidus (DI). In routine cases the patient should not drink any fluids after dinner the day before the test. In cases with more severe polyuria and polydipsia this may be too long a period without fluids and the water deprivation should be begun early (e.g., 6 a.m.) on the morning of the test.
2. Obtain plasma and urine osmolality, serum electrolytes, and a plasma arginine vasopressin (AVP) level at the start of the test.
3. Measure urine volume and osmolality hourly or with each voided urine.
4. Stop the test when body weight decreases by $\geq 3\%$, the patient develops orthostatic blood pressure changes, the urine osmolality reaches a plateau (i.e., $<10\%$ change over two or three consecutive measurements), or the serum $[\text{Na}^+]$ > 145 mmol/L.
5. Obtain plasma and urine osmolality, serum electrolytes, and a plasma AVP level at the end of the test, when the plasma osmolality is elevated, preferably > 300 mOsm/kg H_2O .
6. If the serum $[\text{Na}^+] < 146$ mmol/L or the plasma osmolality < 300 mOsm/kg H_2O when the test is stopped, then consider a short infusion of hypertonic saline (3% NaCl at a rate of 0.1 mL/kg/min for 1–2 h) to reach these endpoints.
7. If hypertonic saline infusion is not required to achieve hyperosmolality, administer AVP (5 U) or desmopressin (1 μg) subcutaneously and continue following urine osmolality and volume for an additional 2 h.

Interpretation

1. An unequivocal urine concentration after AVP/desmopressin ($> 50\%$ increase) indicates central DI and an unequivocal absence of urine concentration ($<10\%$) strongly suggests nephrogenic DI or primary polydipsia.
2. Differentiating between nephrogenic DI and primary polydipsia, as well as for cases in which the increase in urine osmolality after AVP/desmopressin administration is more equivocal (e.g., 10–50%), is best done using the relation between plasma AVP levels and plasma osmolality obtained at the end of the dehydration period and/or hypertonic saline infusion and the relation between plasma AVP levels and urine osmolality under basal conditions (Robertson, 1995; Verbalis, 2012).

VlbR on corticotrophs, which may also contribute to increased systemic AVP levels (Oelkers, 1989).

The recent development of a commercial assay for the C-terminal glycoprotein of the AVP prohormone copeptin offers the possibility of a more stable and easier to measure marker of AVP secretion in response to induced dehydration and hypertonicity. Copeptin is secreted from the posterior pituitary in equimolar amounts as AVP, but is not degraded proteolytically as rapidly as AVP; in addition, its larger size renders measurement by two-site immunoassays possible, in contrast to AVP, which is too small for use of this technique (Morgenthaler et al., 2008). However, the clinical use of copeptin levels as a surrogate marker of AVP secretion during fluid deprivation tests will require standardization of normal and abnormal responses relative to plasma AVP levels.

Clinical manifestations

The characteristic clinical symptoms of DI are the polyuria and polydipsia that result from the underlying impairment of urinary concentrating mechanisms, which have already been covered in the previous section discussing pathophysiology of specific types of DI. Interestingly, patients with DI typically describe a craving for cold water, which appears to quench their thirst

better. Patients with CDI also typically describe a precipitous onset of their polyuria and polydipsia, which simply reflects the fact that urinary concentration can be maintained fairly well until the number of AVP-producing neurons in the hypothalamus decreases to 10–15% of normal, after which plasma AVP levels decrease to the range where urine output increases dramatically (Heinbecker and White, 1941).

However, patients with DI, and particularly those with osmoreceptor dysfunction syndromes, can also present with varying degrees of hyperosmolality and dehydration depending on their overall hydration status. It is therefore important to be aware of the clinical manifestations of hyperosmolality as well. These can be divided into the signs and symptoms produced by dehydration, which are largely cardiovascular, and those caused by the hyperosmolality itself, which are predominantly neurologic and reflect brain dehydration as a result of osmotic water shifts out of the central nervous system (Verbalis, 2012). Cardiovascular manifestations of hypertonic dehydration include hypotension, azotemia, acute tubular necrosis secondary to renal hypoperfusion or rhabdomyolysis, and shock. Neurologic manifestations range from nonspecific symptoms such as irritability and cognitive dysfunction to more severe manifestations of *hypertonic encephalopathy* such as disorientation, decreased level of consciousness,

obtundation, chorea, seizures, coma, focal neurologic deficits, subarachnoid hemorrhage, and cerebral infarction. The severity of symptoms can be roughly correlated with the degree of hyperosmolality, but individual variability is marked and for any single patient the level of serum $[\text{Na}^+]$ at which symptoms will appear cannot be accurately predicted.

Similar to hypo-osmolar syndromes, the length of time over which hyperosmolality develops can markedly affect the clinical symptomatology. Rapid development of severe hyperosmolality is frequently associated with marked neurologic symptoms, whereas gradual development over several days or weeks generally causes milder symptoms. In this case, the brain counteracts osmotic shrinkage by increasing intracellular content of solutes. These include electrolytes such as potassium and a variety of *organic osmolytes*, which were previously called *idiogenic osmoles*; for the most part these are the same organic osmolytes that are lost from the brain during adaptation to hypo-osmolality (Gullans and Verbalis, 1993). The net effect of this process is to protect the brain against excessive shrinkage during sustained hyperosmolality. However, once the brain has adapted by increasing its solute content, rapid correction of the hyperosmolality can produce brain edema, since it takes a finite time (24–48 hours in animal studies) to dissipate the accumulated solutes, and until this process has been completed the brain will accumulate excess water as plasma osmolality is normalized (Verbalis, 2010). This effect is most often seen in dehydrated pediatric patients who can develop seizures with rapid rehydration, but it has been described only rarely in adults.

Therapy

The general goals of treatment of all hyperosmolar disorders are (1) correction of pre-existing water deficits, and (2) reduction in ongoing excessive urinary water losses. The specific therapy required varies with the clinical situation. Awake ambulatory patients with diabetes insipidus and normal thirst have little body water deficit but benefit from relief of the polyuria and polydipsia that disrupt normal activities. In contrast, comatose patients with or without diabetes insipidus are unable to drink in response to thirst, and in these patients progressive hypertonicity may be life-threatening. The established water deficit may be estimated using the following formula (Robinson and Verbalis, 1997):

$$\text{Water deficit} = 0.6 \times \text{premorbid weight} \times [1 - 140 / \text{serum } [\text{Na}^+] \text{ (mmol/L)}]$$

This formula is dependent on several assumptions (total body water is approximately 60% of body weight, no body solute is lost as hypertonicity develops, and the

premorbid $[\text{Na}^+]$ is 140 mmol/L), but nonetheless provides a valid estimate of the approximate total body water deficit. To reduce the risk of central nervous system injury from protracted exposure to severe hypertonicity, the serum osmolality should be lowered to the range of 330 mOsm/kg H_2O within the first 24 hours of therapy. As noted previously, because the organic osmolytes accumulated in the brain during chronic hyperosmolality cannot be immediately dissipated, further correction to a normal osmolality should be spread over the subsequent 1–3 days to avoid producing cerebral edema during treatment, particularly in pediatric patients (Adroque and Madias, 2000a).

It should be remembered that formulae which estimate body water deficits do not take ongoing water losses into account. Consequently, frequent serum electrolyte determinations must be made, and the administration rate of oral water or intravenous 5% dextrose in water should be adjusted accordingly. For example, the estimated water deficit of a 70 kg patient whose serum $[\text{Na}^+]$ is 160 mmol/L would be 5.25 L. In such an individual, administration of water at a rate > 200 mL per hour would be required simply to correct the established deficit over 24 hours, but additional fluid would be needed to keep up with any ongoing losses in a patient with diabetes insipidus until a response to treatment has occurred.

A variety of antidiuretic agents have been used to treat central diabetes insipidus, but desmopressin is the treatment of choice for this disorder. Desmopressin was synthesized as a selective antagonist of AVP V2R, and it is particularly useful therapeutically because it has a much longer half-life than AVP and is devoid of the pressor activity of AVP at vascular V1aR (Robinson, 1976). Desmopressin is generally administered intranasally (5–20 μg every 8–24 hours), but can be given parenterally in acute situations (1–2 μg via the intravenous, intramuscular, or subcutaneous route). For both the intranasal and parenteral preparations, increasing the administered dose generally has the effect of prolonging the duration of antidiuresis rather than increasing its magnitude; consequently, altering the dose can be useful to reduce the required frequency of administration. Synthetic AVP (Pitressin) can also be used to treat central DI, but its use is limited by a much shorter half-life necessitating more frequent dosing or a continuous infusion, and the production of pressor effects due to vasoconstriction.

Nephrogenic diabetes insipidus is more difficult to treat since the kidney is resistant to all AVP-type agents. Limited responses can sometimes be achieved using thiazide diuretics (any drug of the thiazide class may be used with equal potential for benefit). Thiazides cause natriuresis by blocking sodium absorption in the cortical

diluting site; when combined with dietary sodium restriction a modest hypovolemia results, which stimulates isotonic proximal tubular solute reabsorption and diminishes solute delivery to the distal parts of the nephron. Together, these effects diminish free water clearance independently of actions of AVP, thereby decreasing the polyuria of patients with nephrogenic DI (Sands and Bichet, 2006). Monitoring for hypokalemia is necessary and K^+ supplementation is occasionally required. Care must be exercised when treating patients taking lithium with diuretics, since the induced contraction of plasma volume may increase lithium concentrations by increasing proximal tubular absorption and worsen potential toxic effects of the therapy (Grunfeld and Rossier, 2009). Because prostaglandins increase renal medullary blood flow and diminish medullary solute concentration, effects that modestly decrease the interstitial gradient for water reabsorption, drugs that block renal prostaglandin synthesis (e.g., nonsteroidal anti-inflammatory agents) can increase non-AVP-mediated water reabsorption and impair urinary dilution, thereby reducing free water clearance and urine output. Although these agents are somewhat effective in central diabetes insipidus, their main usefulness is as adjunctive therapy in nephrogenic diabetes insipidus, in which more direct antidiuretic therapies are limited.

REFERENCES

- Adroge HJ, Madias NE (2000a). Hyponatremia. *N Engl J Med* 342: 1493–1499.
- Adroge HJ, Madias NE (2000b). Hyponatremia. *N Engl J Med* 342: 1581–1589.
- Ayus JC (1986). Diuretic-induced hyponatremia [editorial]. *Arch Intern Med* 146: 1295–1296.
- Babey M, Kopp P, Robertson GL (2011). Familial forms of diabetes insipidus: clinical and molecular characteristics. *Nat Rev Endocrinol* 7: 701–714.
- Bartter FC, Schwartz WB (1967). The syndrome of inappropriate secretion of antidiuretic hormone. *Am J Med* 42: 790–806.
- Battison C, Andrews PJ, Graham C et al. (2005). Randomized controlled trial on the effect of a 20% mannitol solution and a 75% saline/6% dextran solution on increased intracranial pressure after brain injury. *Crit Care Med* 33: 196–202.
- Baylis PH, Thompson CJ (1988). Osmoregulation of vasopressin secretion and thirst in health and disease. *Clin Endocrinol (Oxf)* 29: 549–576.
- Baylis PH, Thompson CJ (2001). Diabetes insipidus and hyperosmolar syndromes. In: KL Becker (Ed.), *Principles and Practice of Endocrinology and Metabolism*. 3rd edn. Lippincott Williams Wilkins, Philadelphia, pp. 285–293.
- Berl T (2008). Impact of solute intake on urine flow and water excretion. *J Am Soc Nephrol* 19: 1076–1078.
- Berl T, Quittnat-Pelletier F, Verbalis JG et al. (2010). Oral tolvaptan is safe and effective in chronic hyponatremia. *J Am Soc Nephrol* 21: 705–712.
- Bissram M, Scott FD, Liu L et al. (2007). Risk factors for symptomatic hyponatraemia: the role of pre-existing asymptomatic hyponatraemia. *Intern Med J* 37: 149–155.
- de Castro JM (1988). A microregulatory analysis of spontaneous fluid intake in humans: evidence that the amount of liquid ingested and its timing is mainly governed by feeding. *Physiol Behav* 3: 705–714.
- Dcaux G (2009). The syndrome of inappropriate secretion of antidiuretic hormone (SIADH). *Semin Nephrol* 29: 239–256.
- Dcaux G, Genette F (1981). Urea for long-term treatment of syndrome of inappropriate secretion of antidiuretic hormone. *Br Med J (Clin Res Ed)* 283: 1081–1083.
- Dcaux G, Waterlot Y, Genette F et al. (1981). Treatment of the syndrome of inappropriate secretion of antidiuretic hormone with furosemide. *N Engl J Med* 304: 329–330.
- Denton D (1982). *The Hunger for Salt: An Anthropological Physiological and Medical Analysis*. Springer-Verlag, Berlin.
- Ellison DH, Berl T (2007). Clinical practice. The syndrome of inappropriate antidiuresis. *N Engl J Med* 356: 2064–2072.
- Fitzsimons JT (1992). Physiology and pathophysiology of thirst and sodium appetite. In: DW Seldin, G Giebisch (Eds.), *The Kidney. Physiology and Pathophysiology*. Raven Press, New York, pp. 1615–1648.
- Fujiwara TM, Bichet DG (2005). Molecular Biology of Hereditary Diabetes Insipidus. *J Am Soc Nephrol* 16: 2836–2846.
- Furst H, Hallows KR, Post J et al. (2000). The urine/plasma electrolyte ratio: a predictive guide to water restriction. *Am J Med Sci* 319: 240–244.
- Goldschmidt MA, Iliescu EA (2000). DDAVP to prevent rapid correction in hyponatremia. *Clin Nephrol* 53: 226–229.
- Greenberg A, Verbalis JG (2006). Vasopressin receptor antagonists. *Kidney Int* 69: 2124–2130.
- Grunfeld JP, Rossier BC (2009). Lithium nephrotoxicity revisited. *Nat Rev Nephrol* 5: 270–276.
- Gullans SR, Verbalis JG (1993). Control of brain volume during hyperosmolar and hypoosmolar conditions. *Annu Rev Med* 44: 289–301.
- Heinbecker P, White HL (1941). Hypothalamico-hypophyseal system and its relation to water balance in the dog. *Am J Physiol* 133: 582–593.
- Hew-Butler T, Ayus JC, Kipps C et al. (2008). Statement of the Second International Exercise-Associated Hyponatremia Consensus Development Conference, New Zealand, 2007. *Clin J Sport Med* 18: 111–121.
- Ishikawa S, Fujita N, Fujisawa G et al. (1996). Involvement of arginine vasopressin and renal sodium handling in pathogenesis of hyponatremia in elderly patients. *Endocr J* 43: 101–108.
- Johnson AK, Buggy J (1978). Periventricular preoptic-hypothalamus is vital for thirst and normal water economy. *Am J Physiol* 234: R122–R129.
- Khanna A (2006). Acquired nephrogenic diabetes insipidus. *Semin Nephrol* 26: 244–248.
- Kirchner KA, Stein JH (1994). Sodium metabolism. In: RG Narins (Ed.), *Clinical Disorders of Fluid and*

- Electrolyte Metabolsim. 5th edn. McGraw-Hill, New York, pp. 45–80.
- Knepper MA (1997). Molecular physiology of urinary concentrating mechanism: regulation of aquaporin water channels by vasopressin. *Am J Physiol* 272: F3–F12.
- Knepper MA (1998). Long-term regulation of urinary concentrating capacity. *Am J Physiol* 275: F332–F333.
- Li-Ng M, Verbalis JG (2010). Conivaptan: evidence supporting its therapeutic use in hyponatremia. *Core Evid* 4: 83–92.
- Morgenthaler NG, Struck J, Jochberger S et al. (2008). Copeptin: clinical use of a new biomarker. *Trends Endocrinol Metab* 19: 43–49.
- Nielsen S, Frokiaer J, Marples D et al. (2002). Aquaporins in the kidney: from molecules to medicine. *Physiol Rev* 82: 205–244.
- Oelkers W (1989). Hyponatremia and inappropriate secretion of vasopressin (antidiuretic hormone) in patients with hypopituitarism. *N Engl J Med* 321: 492–496.
- Ohnishi A, Orita Y, Okahara R et al. (1993). Potent aquaretic agent. A novel nonpeptide selective vasopressin 2 antagonist (OPC-31260) in men. *J Clin Invest* 92: 2653–2659.
- Orth DN, Kovacs WJ (1998). The adrenal cortex. In: JD Wilson, DW Foster, HM Kronenberg et al. (Eds.), *Williams Textbook of Endocrinology*. 9th edn. WB Saunders, Philadelphia, pp. 517–664.
- Otsuka Pharmaceutical Co., Ltd., Tokyo (2009). Samsca (tolvaptan) prescribing information.
- Oya S, Tsutsumi K, Ueki K et al. (2001). Reinduction of hyponatremia to treat central pontine myelinolysis. *Neurology* 57: 1931–1932.
- Phillips PA, Rolls BJ, Ledingham JG et al. (1985). Angiotensin II-induced thirst and vasopressin release in man. *Clin Sci (Lond)* 68: 669–674.
- Reeves WB, Andreoli TE (2001). Tubular sodium transport. In: RW Schrier (Ed.), *Diseases of the Kidney and Urinary Tract*. 7th edn. Lippincott Williams Wilkins, Philadelphia, pp. 135–175.
- Renneboog B, Musch W, Vandemergel X et al. (2006). Mild chronic hyponatremia is associated with falls unsteadiness and attention deficits. *Am J Med* 119: 71.
- Robertson GL (1976). The regulation of vasopressin function in health and disease. *Recent Prog Horm Res* 33: 333–385.
- Robertson GL (1983). Thirst and vasopressin function in normal and disordered states of water balance. *J Lab Clin Med* 101: 351–371.
- Robertson GL (1995). Diabetes insipidus. *Endocrinol Metab Clin North Am* 24: 549–572.
- Robertson GL (2006). Regulation of arginine vasopressin in the syndrome of inappropriate antidiuresis. *Am J Med* 119 (7 Suppl 1): S36–S42.
- Robinson AG (1976). DDAVP in the treatment of central diabetes insipidus. *N Engl J Med* 294: 507–511.
- Robinson AG, Verbalis JG (1997). Diabetes insipidus. *Curr Ther Endocrinol Metab* 6: 1–7.
- Robinson AG, Verbalis JG (2011). Posterior pituitary. In: S Melmed, KS Polonsky, P Reed Larsen et al. (Eds.), *Williams Textbook of Endocrinology*. 12th edn. WB Saunders, Philadelphia, pp. 291–323.
- Sands JM, Bichet DG (2006). Nephrogenic diabetes insipidus. *Ann Intern Med* 144: 186–194.
- Schrier RW, Gross P, Gheorghiade M et al. (2006). Tolvaptan a selective oral vasopressin V2-receptor antagonist for hyponatremia. *N Engl J Med* 355: 2099–2112.
- Schwartz WB, Bennett S, Curelop S et al. (1957). A syndrome of renal sodium loss and hyponatremia probably resulting from inappropriate secretion of antidiuretic hormone. *Am J Med* 23: 529–542.
- Singer I, Rotenberg D (1973). Demeclocycline-induced nephrogenic diabetes insipidus. In-vivo and in-vitro studies. *Ann Intern Med* 79: 679–683.
- Soupart A, Penninckx R, Crenier L et al. (1994). Prevention of brain demyelination in rats after excessive correction of chronic hyponatremia by serum sodium lowering. *Kidney Int* 45: 193–200.
- Steele A, Gowrishankar M, Abrahamson S et al. (1997). Postoperative hyponatremia despite near-isotonic saline infusion: a phenomenon of desalination. [see comments] *Ann Intern Med* 126: 20–25.
- Sterns RH, Riggs JE, Schochet Jr SS (1986). Osmotic demyelination syndrome following correction of hyponatremia. *N Engl J Med* 314: 1535–1542.
- Sterns RH, Cappuccio JD, Silver SM et al. (1994). Neurologic sequelae after treatment of severe hyponatremia: a multicenter perspective. *J Am Soc Nephrol* 4: 1522–1530.
- Sterns RH, Nigwekar SU, Hix JK (2009). The treatment of hyponatremia. *Semin Nephrol* 29: 282–299.
- Stricker EM, Verbalis JG (2013). Water and salt intake and body fluid homeostasis. In: LR Squire, D Berg, FE Bloom et al. (Eds.), *Fundamental Neuroscience*. 4th edn. Academic Press, Waltham MA, pp. 783–797.
- Thibonnier M, Conarty DM, Preston JA et al. (1998). Molecular pharmacology of human vasopressin receptors. *Adv Exp Med Biol* 449: 251–276.
- Thibonnier M, Coles P, Thibonnier A et al. (2001). The basic and clinical pharmacology of nonpeptide vasopressin receptor antagonists. *Annu Rev Pharmacol Toxicol* 41: 175–202.
- Thompson CJ, Bland J, Burd J et al. (1986). The osmotic thresholds for thirst and vasopressin release are similar in healthy man. *Clin Sci (Lond)* 71: 651–656.
- Verbalis JG (1997a). Body sodium and extracellular fluid volume. In: RL Jamison, R Wilkinson (Eds.), *Nephrology*. Chapman Hall Medical, London, pp. 95–101.
- Verbalis JG (1997b). Body water and osmolality. In: RL Jamison, R Wilkinson (Eds.), *Nephrology*. Chapman Hall Medical, London, pp. 89–94.
- Verbalis JG (2009). Hyponatremia and hypo-osmolar disorders. In: A Greenberg (Ed.), *Primer on Kidney Diseases*. 5th edn. Saunders Elsevier, Philadelphia, pp. 52–59.
- Verbalis JG (2010). Brain volume regulation in response to changes in osmolality. *Neuroscience* 168: 862–870.
- Verbalis JG (2012). Disorders of water balance. In: MW Taal, GM Chertow, PA Marsden et al. (Eds.), *Brenner Rector's The Kidney*. 9th edn. Elsevier, Philadelphia, pp. 540–594.
- Verbalis JG (2013). The syndrome of inappropriate antidiuretic hormone secretion and other hypoosmolar

- disorders. In: TM Coffman, RJ Falk, BA Molitoris et al. (Eds.), *Schrier's Diseases of the Kidney*. 9th edn. Lippincott Williams Wilkins, Philadelphia, pp. 2012–2054.
- Verbalis JG, Goldsmith SR, Greenberg A et al. (2013). Diagnosis, evaluation, and treatment of hyponatremia: expert panel recommendations. *Am J Med* 126 (10 Suppl 1): S1–S42.
- Wilkins L, Richter CP (1940). A great craving for salt by a child with cortico-adrenal insufficiency. *J Am Med Assoc* 114: 866–868.
- Zeltser D, Rosansky S, van Rensburg H et al. (2007). Assessment of the efficacy and safety of intravenous conivaptan in euvolemic and hypervolemic hyponatremia. *Am J Nephrol* 27: 447–457.